

Impact of Non-Uniform Constellations on MLSE Demodulation Performance for PCS-16QAM under Bandwidth Limitation

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Abstract We investigated the impact of the prior probability term in MLSE branch metrics and demonstrated a GMI improvement of up to 9% over the case without MLSE by optimizing the prior probability term weight through transmission experiments using PCS-16QAM signals under severe bandwidth limitation. ©2026 The Author(s)

Introduction

Digital coherent transceivers can achieve high-capacity optical transmission by exploiting optical properties, including phase and polarization together with advanced signal processing [1–4]. In the near future, high-capacity optical transmission exceeding 2 Tbps per wavelength will be required. With the growing demand for higher-capacity transmission, optical signal-to-noise ratio (OSNR) tolerance has become a critical issue in optical communication systems. To address this issue, probabilistic constellation shaping (PCS) has recently been used [5,6], as it approaches the Shannon limit by optimizing the symbol probability distribution, and thereby improves transmission efficiency. Moreover, increasing the baud rate is also essential for improving the data rate, but signal degradation caused by bandwidth limitations (BWLs) of optical and electrical devices, which induce inter-symbol interference (ISI), poses a significant challenge. Maximum likelihood sequence estimation (MLSE) is an effective signal processing method for BWLs [7–10].

For MLSE under PCS coding, the prior probability term should be incorporated into the branch metrics [11]. However, in optical transmission systems under severe BWLs, it remains unclear to what extent the prior probability term should be incorporated into the branch metrics.

In this paper, we investigated the impact of the weighting of the prior probability term of PCS signals on the branch metric calculation in MLSE and the GMI optimization under strong BWLs. Through transmission experiments on PCS-16QAM signals with a transceiver bandwidth limited mainly by a 23-GHz optical modulator, while varying the entropy, baud rate, and OSNR conditions, we revealed that the prior probability term in the branch metrics should be scaled to 0.4–0.6

times its theoretical value to maximize the GMI, deviating from the theoretically expected scaling factor of 1, and the GMI was improved by up to 9 % compared with the case without MLSE. This deviation was partly attributed to the non-Gaussian distribution of the residual noise, in which the noise was concentrated at small amplitudes, causing the prior probability term to be over-weighted under the Gaussian assumption.

Principle of MLSE under PCS coding

Fig. 1 shows the block diagram of MLSE under PCS coding. A low-pass filter (LPF) attenuates the high-frequency component of the received sequence y , while the desired impulse response filter (DIRF) $\hat{H}$ represents the estimated transfer function which emulates the channel response including BWLs for candidate sequences x . Both filters are linear finite impulse response filters and are updated by a decision-directed least mean square algorithm. In the case of the PCS signals, the branch metrics γ at timing i are calculated as

$$\gamma_i = (y_i - \sum_k h_k x_{i-k})^2 - 2\alpha\sigma^2 \ln P(x_i). \quad (1)$$

where y_i denotes the received sequence y , x_i denotes the symbol of the candidate sequence, h_k denotes the coefficient of $\hat{H}$, σ^2 denotes the noise variance of the received signal after the LPF, and α denotes the coefficient. The first term on the right-hand side is referred to as the likelihood term, which measures the similarity between the received sequence and the candidate sequence under BWLs. The second term on the right-hand side is referred to as the prior probability term. Although the coefficient α is theoretically equal to 1, it should be appropriately adjusted according to the transmission conditions.

The branch metrics are then fed into the Viterbi algorithm (VA) decoder to estimate the maximum-likelihood symbol sequence, followed by log-likelihood ratio (LLR) calculation.

Experiment

1. Experimental setup

The experimental setup is shown in Fig. 2(a). In

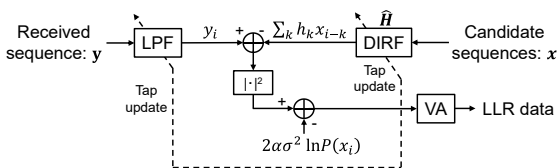


Fig. 1: Block diagram of MLSE under PCS coding.

these experiments, PCS-16QAM optical signals were generated using a 32-GHz bandwidth, 96-GSa/s arbitrary waveform generator (AWG), 25-GHz bandwidth driver amplifiers, and a 23-GHz bandwidth polarization division multiplexing IQ modulator (PDM-IQM). The baud rate was set to 64, 70 and 76 Gbaud. The PCS signal was generated using the constant composition distribution matching (CCDM) method [5], and the entropy was set to 7.8, 6.8, and 6.2 bits/symbol. The electrical signals were passed through a Nyquist filter with a roll-off factor of 0.1. The wavelength of the integrable tunable laser assembly (ITLA) generating optical signals was set to 1555.344 nm. The optical spectrum at the PDM-IQM output was degraded because of the analog BWs of the transmitter including the AWG, driver amplifiers, and PDM-IQM. The received signals were amplified by an erbium-doped fibre amplifier (EDFA), and the OSNR was controlled by adding amplified spontaneous emission (ASE) noise and set to 22 and 26 dB. The signals were then filtered by a 100-GHz optical bandpass filter (OBPF) and adjusted to 1 dBm by a variable optical attenuator (VOA). The power of the local oscillator (LO) light was adjusted to 14.5 dBm. The received signal and LO light were coupled at a 90° optical hybrid and converted into electrical signals using 70-GHz bandwidth balanced photo detectors (BPDs). The received electrical signals were sampled using a 36-GHz bandwidth 80-GSa/s DSO. Fig. 2(b) shows the configuration of offline DSP, which included resampling, frequency-domain (FD) 8×2 MIMO adaptive equalization (AEQ) [12] with a pilot-based digital phase-locked loop, carrier phase recovery (CPR), and MLSE [7]. MLSE was applied to four lanes (XI, XQ, YI, YQ) independently. The tap lengths of the LPF and DIRF of MLSE were both set to three. The GMI was then calculated from the LLR.

In MLSE, the branch metrics were calculated using Eq. (1). Transmission experiments were conducted while varying α from 0 to 1.2, and the GMI was evaluated. Prior to the VA and LLR

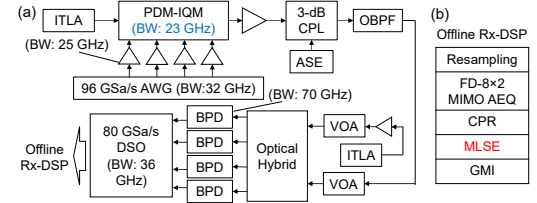


Fig. 2. (a) Experimental setup. (b) Configuration of offline digital signal processing.

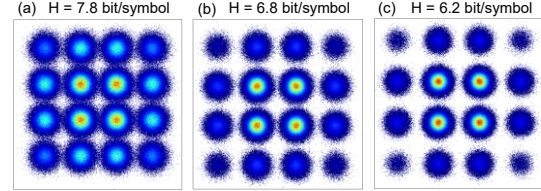


Fig. 3. Constellation of PCS signal with entropy of (a) 7.8 bit/symbol, (b) 6.8 bit/symbol, and (c) 6.2 bit/symbol at OSNR = 22 dB and baud rate = 64 Gbaud.

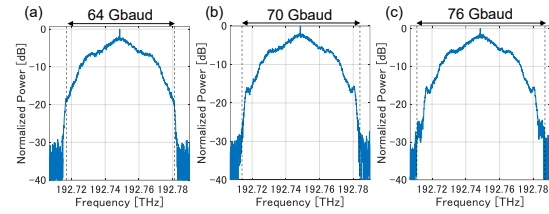


Fig. 4. Optical output spectrum at a baud rate of (a) 64 Gbaud, (b) 70 Gbaud, and (c) 76 Gbaud with entropy = 6.8 bit/symbol.

calculations, 200,000 training symbols were used for tap convergence and for estimating the noise variance after the LPF as,

$$\sigma_{est}^2 = \frac{1}{N} \sum_i^N (y_i - \sum_k h_k X_{i-k})^2. \quad (2)$$

where X_i represents the training symbols and N represents the averaging number.

Fig. 3 shows the simulated constellations of the PCS signals for each entropy when the OSNR and baud rate were set to 22 dB and 64 Gbaud, respectively. As the entropy decreases, the occurrence probability of the outer symbols also decreases. Fig. 4 shows the optical output spectrum of each baud rate. The higher the baud rate, the more severe the BWs become.

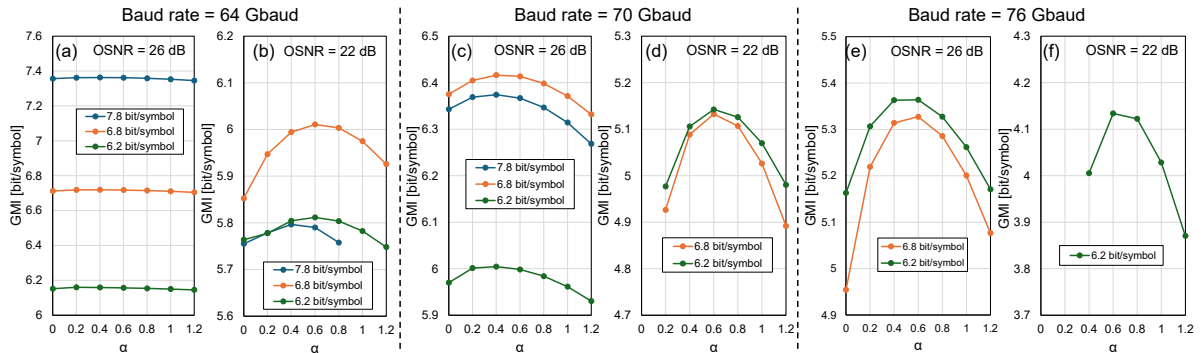
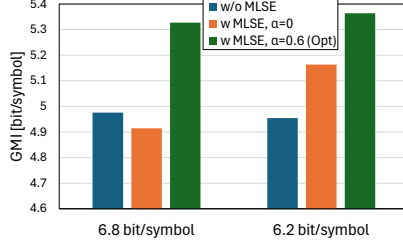


Fig. 5. GMI as a function of α at OSNR = (a, c, e) 26 dB and (b, d, f) 22 dB, and baud rate = (a, b) 64 Gbaud, (c, d) 70 Gbaud and (e, f) 76 Gbaud.

Tab. 1: Optimal α at OSNR = (a) 26dB and (b) 22dB

(a) OSNR = 26 dB			(b) OSNR = 22 dB			
Entropy (bit/symbol)	Baud rate (Gbaud)		Entropy (bit/symbol)	Baud rate (Gbaud)		
	70	76		64	70	76
7.8	0.4	—	7.8	0.4	—	—
6.8	0.4	0.6	6.8	0.6	0.6	—
6.2	0.4	0.6	6.2	0.6	0.6	0.6

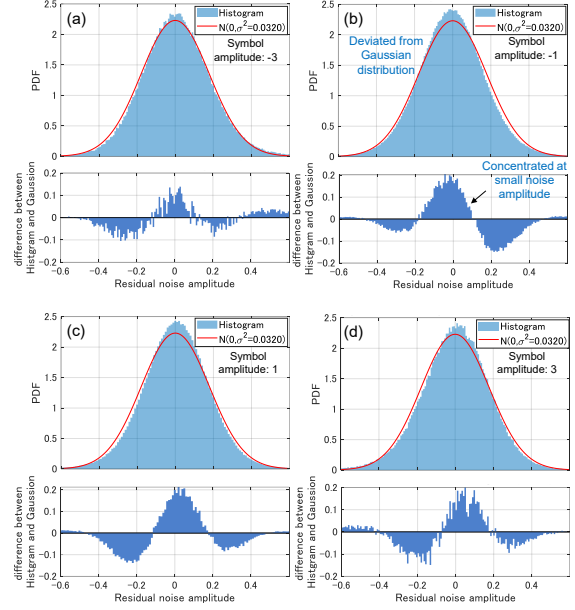
**Fig. 6.** Comparison of GMI with MLSE ($\alpha = 0$ and optimal) and without MLSE for entropies of 6.8 and 6.2 bit/symbol at OSNR = 26 dB and baud rate = 76 Gbaud.

2. Experimental result

Fig. 5 shows the GMI as a function of α for OSNRs of 26 and 22 dB, baud rates of 64, 70, and 76 Gbaud, and entropies of 7.8, 6.8, and 6.2 bit/symbol. Note that some data points are missing because the MLSE taps failed to converge, preventing reliable measurement. When the OSNR was high and the baud rate was low, the GMI was nearly independent of α , as shown in Fig. 5(a). However, as the OSNR decreased and the baud rate increased, the dependence of the GMI on α became stronger, as shown in Fig. 5(b) to (f). This is because, under low OSNR conditions, the noise variance σ^2 increases, leading to a larger contribution of the prior probability term. In addition, under high baud rate conditions, the received bandwidth widens, increasing the noise power even at the same OSNR. Moreover, since the system bandwidth is fixed, a higher baud rate leads to stronger BWLs relative to the signal bandwidth, and residual ISI further increases σ^2 . As σ^2 increases, the prior probability term $2\alpha\sigma^2 \ln P(x_i)$ in Eq. (1) grows in weight relative to the likelihood term, making the candidate sequence selected by the VA decoder more sensitive to the value of α .

Table 1 summarizes the optimal α under various conditions. As in Fig. 5, some points were unavailable for the same reason. The optimal α that maximizes the GMI was found in the range of 0.4–0.6, deviating significantly from the theoretical value of 1, which is expected when the residual noise follows a Gaussian distribution.

Fig. 6 shows the GMI comparison of with and without MLSE. By incorporating the prior probability term with optimal α , which appropriately

**Fig. 7.** Residual noise amplitude histogram, Gaussian distribution with the noise variance and their difference at amplitude of (a) -3, (b) -1, (c) 1, and (d) 3 in XI lane, at OSNR = 26 dB, entropy = 6.8 bit/symbol, and baud rate = 76 Gbaud.

reflects the probability distribution of the PCS signals, into the MLSE, the GMI was improved by approximately 7% and 9% for entropies of 6.8 and 6.2 bit/symbol, respectively, compared with the case without MLSE.

Fig. 7 shows the histogram of the residual noise amplitude estimated from the training symbols in the XI lane, together with a Gaussian distribution having the noise variance (calculated by Eq. (2)) and the difference between the histogram and the Gaussian distribution at each amplitude. The OSNR, entropy, and baud rate were set to 26 dB, 6.8 bit/symbol, and 76 Gbaud, respectively. As shown in Fig. 7, the histogram deviated from a Gaussian distribution, particularly at inner amplitudes. Hence, the Gaussian noise assumption is no longer optimal. Since the noise was concentrated at small amplitudes (−0.4 to 0.4), the prior probability term became overweighted under the Gaussian assumption, and α should therefore be set to less than 1 as compensation.

Conclusions

We investigated the impact of non-uniform constellations on the MLSE branch metric. Through transmission experiments using PCS-16QAM under severe BWLs, we demonstrated that scaling the prior probability term by a factor of 0.4–0.6 relative to its theoretical Gaussian-based value maximized the GMI, yielding up to a 9% improvement over the case without MLSE. This deviation was partly attributed to the sharper-than-Gaussian peak of the residual noise distribution.

References

- [1] F. Pittalà, "High-Capacity and High-Spectral Efficiency Transmission Systems for 1.6 Tb/s and Beyond," in Proceedings of Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2024, Tu3E.4. DOI: [10.1364/OFC.2024.Tu3E.4](https://doi.org/10.1364/OFC.2024.Tu3E.4)
- [2] G. Huang, Y. Nakamura, H. Nakashima, and T. Hoshida, "240GBd-16QAM Single-Carrier Coherent Transmission over 120km SSMF for a Bandwidth Limited System with 1sps Speed and Simple DSP," in Proceedings of Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2024, M2H.2. DOI: [10.1364/OFC.2024.M2H.2](https://doi.org/10.1364/OFC.2024.M2H.2)
- [3] M. Nakamura, H. Taniguchi, S. Yamamoto, F. Hamaoka, M. Nagatani, T. Jyo, M. Mutoh, Y. Shiratori, H. Wakita, T. Kobayashi, H. Takahashi, and Y. Miyamoto, "Beyond 200-GBd QAM Signal Detection Based on Trellis-path-limited Sequence Estimation Supporting Soft-decision Forward Error Correction", in Proceedings of Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2023, M1F.2. DOI: [10.1364/OFC.2023.M1F.2](https://doi.org/10.1364/OFC.2023.M1F.2)
- [4] A. Busson, H. Bissessur, F. Hedaraly, A. Calsat, P. Plantady, "Real Time C-Band Unrepeated Transmission of 36.4 Tb/S in Pcs-64Qam and 32 Tb/S in Pcs-16Qam Over 368 Km and 407 Km Respectively," in Proceedings of European Conference on Optical Communication (ECOC), Copenhagen, Denmark, 2025, Th3C.1. DOI: [10.1109/ECOC66593.2025.11263270](https://doi.org/10.1109/ECOC66593.2025.11263270)
- [5] P. Schulte and G. Böcherer, "Constant Composition Distribution Matching," IEEE Transactions on Information Theory, vol. 62, no. 1, pp. 430–434, 2016. DOI: [10.1109/TIT.2015.2499181](https://doi.org/10.1109/TIT.2015.2499181)
- [6] T. Yoshida, M. Karlsson, and E. Agrell, "Hierarchical Distribution Matching for Probabilistically Shaped Coded Modulation," Journal of Lightwave Technology, vol. 37, no. 6, pp. 1579–1589, 2019. DOI: [10.1109/JLT.2019.2891680](https://doi.org/10.1109/JLT.2019.2891680)
- [7] H. Taniguchi, S. Yamamoto, M. Nakamura, A. Masuda, Y. Kisaka, S. Kanazawa, and H. Nakamura, "High-Capacity IM-DD Transmission in O-Band Using Advanced Maximum Likelihood Sequence Estimation Methods," Journal of Lightwave Technology, vol. 41, no. 11, pp. 3372–3381, 2023. DOI: [10.1109/JLT.2023.3253792](https://doi.org/10.1109/JLT.2023.3253792)
- [8] S. Yamamoto, H. Taniguchi, M. Nakamura, and Y. Kisaka, "High-speed short-reach O-band transmission using advanced DSP techniques," in Proceedings of Asia Communications and Photonics Conference (ACP) and International Conference on Information Photonics and Optical Communications (IPOC), 2020, S4H.1. DOI: [10.1364/ACPC.2020.S4H.1](https://doi.org/10.1364/ACPC.2020.S4H.1)
- [9] H. Zhang, "Power Efficient Coherent Detection for Short-Reach System," in Proceedings of Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2023, M1E.1. DOI: [10.1364/OFC.2023.M1E.1](https://doi.org/10.1364/OFC.2023.M1E.1)
- [10] E. Pincemin, N. Genay, B. L. Guyader, and J. Wang, "800G WDM Transmission with 140-GBaud PCS-16QAM Transceiver Prototype and 150-GHz Channel Spacing on 2000-km SSMF with EDFA Only," in Proceedings of Optical Fiber Communications Conference and Exhibition (OFC), Los Angeles, CA, USA, 2025, Tu3I.2. DOI: [10.1364/OFC.2025.Tu3I.2](https://doi.org/10.1364/OFC.2025.Tu3I.2)
- [11] X. Jia, Y. Li, Y. Liu, H. Zhou, J. Li, M. Luo, Z. He, C. Yang, J. Qiu, Y. Zuo, W. Li, X. Hong, H. Guo, J. Lv, and J. Wu, "Performance Investigation of the Probability-Aided Maximum Likelihood Sequence Detector for a Probabilistic Shaped 16-QAM System", Optics Express, vol. 30, Issue 9, pp. 14421–14431, 2022. DOI: [10.1364/OE.453610](https://doi.org/10.1364/OE.453610)
- [12] M. Nakamura, T. Kobayashi, F. Hamaoka, and Y. Miyamoto, "High Information Rate of 128-GBaud 1.8-Tb/s and 64-GBaud 1.03-Tb/s Signal Generation and Detection Using Frequency-Domain 8×2 MIMO Equalization," in Proceedings of Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2022, M3H.1. DOI: [10.1364/OFC.2022.M3H.1](https://doi.org/10.1364/OFC.2022.M3H.1)